

Evaluation of Instruction-Following of Language Models, Especially with a Multilingual Focus: A Literature Survey

Introduction

Instruction-following capability in large language models (LLMs) is a critical dimension of their utility, reflecting how effectively these models can comprehend and execute user directives across diverse tasks. As LLMs are increasingly deployed in multilingual and multicultural settings, evaluating their instruction adherence in multiple languages becomes essential. Multilingual instruction-following evaluation not only ensures equitable AI performance across languages but also addresses challenges arising from linguistic diversity, cultural nuances, and resource disparities. This survey synthesizes recent advances in benchmarks, methodologies, and model developments focused on instruction-following evaluation, with an emphasis on multilingual contexts. The reviewed literature spans general-purpose LLMs, domain-specific models, reasoning-focused frameworks, and audio or code generation modalities, highlighting progress, persistent challenges, and promising directions.

Benchmarks and Methodologies for Instruction-Following Evaluation

Foundational and Extended Instruction-Following Benchmarks

A foundational contribution to instruction-following evaluation is the Instruction-Following Eval (IFEval) benchmark introduced by Zhou et al. (2023), which emphasizes *verifiable instructions*—measurable and specific directives such as word count or keyword inclusion. IFEval’s reproducible framework addresses shortcomings of costly human evaluations and biased LLM-based auto-evaluation, providing a standardized means to assess instruction adherence. Building on this, Kovalevskyi (2024) proposed IFEval-Extended, which dynamically generates thousands of unique prompts from base templates to reduce overfitting and increase evaluation diversity. This extension introduces more granular instruction categories, including language structure and response formatting, revealing that while models like GPT-4o and LLaMA 3 variants excel at simple instructions, they struggle with complex, multi-constraint tasks.

LexInstructEval (Ren et al., 2025) further advances evaluation by focusing on fine-grained lexical instruction following through a formal, rule-based grammar that decomposes instructions into canonical triplets. This framework enables systematic, objective assessment beyond subjective human judgments or biased automated systems. Although these benchmarks primarily target English or general instruction-following, their design principles suggest adaptability to multilingual contexts.

Multilingual Instruction-Following Benchmarks

Recognizing the need for multilingual assessment, M-IFEval (Dussolle et al., 2025) extends IFEval to French, Japanese, and Spanish, revealing significant performance variability across languages and instruction types, underscoring the necessity of multilingual benchmarks for accurate evaluation. Similarly, Marco-Bench-MIF (Zeng et al., 2025) scales instruction-following evaluation to 30 languages with cultural and linguistic adaptations beyond direct translations. Their findings highlight substantial performance gaps between high- and low-resource languages, model scale benefits, and the underestimation of accuracy by machine-translated data compared to localized datasets.

XIFBench (Li et al., 2025) introduces a multilingual benchmark with 558 instructions spanning six languages and diverse constraint categories, employing cultural accessibility annotations and constraint-level translation validation. The study provides nuanced insights into how language resource levels, instruction complexity, and cultural factors influence instruction adherence.

Multi-IF (He et al., 2024) addresses multi-turn and multilingual instruction following by extending IFEval with multi-turn conversations across eight languages. Their results indicate degraded performance with increasing interaction turns and higher error rates in non-Latin script languages, emphasizing challenges in multilingual conversational instruction adherence.

Evaluation Metrics and Robustness Considerations

Sakai et al. (2024) highlight the importance of accounting for score variance across instruction templates in multilingual settings by proposing cross-lingual datasets with multiple templates per task and introducing the Sharpe score metric. This approach promotes fairer evaluations by capturing variability inherent in natural language instructions.

Liu et al. (2025) focus on robustness against multilingual typographical errors, introducing MulTypo, a realistic typo generation method. Their evaluation reveals that instruction tuning improves performance on clean inputs but may increase vulnerability to noise, with robustness varying by language resource level and translation direction. This underscores the need for noise-aware training and evaluation in multilingual instruction-following.

Zeng et al. (2023) contribute LLMBAR, a meta-evaluation benchmark that rigorously assesses LLM evaluators' ability to judge instruction-following quality. Although primarily English-focused, LLMBAR's design addresses challenges such as negation and lexical constraints relevant to multilingual and complex instructions.

Advances in Multilingual Instruction-Following Models and Training Strategies

Instruction Tuning and Cross-Lingual Transfer

Lai and Nissim (2024) propose mCoT, a multilingual chain-of-thought instruction tuning approach that enhances reasoning consistency across eleven languages, particularly benefitting low-resource languages. Their 7B parameter model achieves strong multilingual performance, demonstrating the effectiveness of multilingual instruction tuning for reasoning tasks.

Zhang et al. (2023) introduce BayLing, an LLM fine-tuned via interactive translation instructions to transfer English instruction-following capabilities to non-English languages, notably Chinese. BayLing achieves translation performance near GPT-4 and robust instruction adherence in multi-turn interactions, illustrating a practical method for cross-lingual knowledge transfer without extensive language-specific pretraining.

Zhang et al. (2023) also propose PLUG, leveraging a high-resource pivot language (English) to improve instruction-following in lower-resource languages. By processing instructions through the pivot language before generating target language responses, PLUG achieves a 29% average performance improvement across four languages, highlighting the utility of pivot-based cross-lingual generation.

Ye et al. (2025) present LangGPS, a data pre-selection framework based on language separability in embedding spaces to guide multilingual instruction tuning. Their approach balances samples with high and low language separability to improve both language boundary clarity and cross-lingual alignment, enhancing instruction-following across 22 languages.

Bu et al. (2025) propose AlignX, a two-stage framework combining multilingual semantic representation alignment with instruction fine-tuning to improve instruction adherence in non-dominant languages. AlignX demonstrates improved cross-lingual generation and understanding, addressing suboptimal multilingual model performance.

Shimabucoro et al. (2025) analyze cross-lingual transfer dynamics in multilingual instruction tuning, revealing that effective transfer depends on complex interactions among training variables. Their insights guide optimization strategies for enhancing multilingual instruction-following in large models.

Domain-Specific and Low-Resource Language Instruction Following

Zhou et al. (2025) introduce PH-LLM, a suite of multilingual LLMs trained on 29 languages for public health surveillance tasks. Their instruction tuning with quantized adapters yields superior instruction-following performance in both English and multilingual tasks compared to leading models, underscoring the importance of domain-specific multilingual training.

Maria (2025) presents Compass-v3, a Mixture-of-Experts model tailored for multilingual e-commerce in Southeast Asia, trained on 12 trillion tokens including low-resource languages. Using Optimal-Transport Direct Preference Optimization, Compass-v3 excels in instruction adherence on commerce-specific tasks and demonstrates strong multilingual capabilities in real-world deployment.

Chitale et al. (2025) develop Updesh, a large-scale synthetic instruction dataset for 13 Indian languages, grounded in culturally relevant Wikipedia content. Models fine-tuned on Updesh show notable improvements in multilingual natural language understanding and generation, emphasizing the value of culturally aware data for instruction-following.

Toleu et al. (2025) focus on Kazakh text simplification, a low-resource, morphologically rich language. Their instruction-tuned model, KazSim, outperforms baselines in both automatic and human evaluations, revealing challenges and strategies for instruction-following in morphologically complex languages.

Khade et al. (2024) explore LoRA PEFT fine-tuning on multilingual models targeting Marathi, a low-resource language. Despite metric declines post-fine-tuning, manual evaluation indicates improved instruction adherence, highlighting evaluation challenges and the need for high-quality native datasets.

Kulkarni et al. (2025) propose a hybrid framework combining pretrained multilingual masked language models with parameter-efficient fine-tuning and cross-lingual retrieval to improve translation and instruction-following in low-resource languages, promoting linguistic diversity and inclusivity.

Manakul et al. (2024) address multilingual instruction-following in audio language models, proposing Typhoon-Audio with balanced English and Thai training data. Their approach significantly improves instruction adherence in Thai while maintaining English performance, highlighting the importance of balanced multilingual audio training.

Multimodal and Specialized Instruction-Following Evaluation

Gao et al. (2025) introduce IFEval-Audio, a benchmark for audio-based LLMs that evaluates adherence to textual instructions derived from audio inputs across multiple structural dimensions. Their findings reveal challenges in maintaining instruction-following performance when integrating audio modalities, suggesting avenues for multilingual and multimodal evaluation.

Yang et al. (2025) propose IFEvalCode, a benchmark assessing instruction-following in code LLMs across seven programming languages with queries in Chinese and English. The study shows that while correctness is often achieved, instruction adherence, particularly regarding style and structure, remains a challenge, with closed-source models outperforming open-source ones.

Wang et al. (2025) develop CodeIF-Bench, focusing on multi-turn interactive code generation instruction-following. Their evaluation highlights the influence of context and history on adherence, providing insights transferable to multilingual instruction-following evaluation.

Sun et al. (2025) evaluate fine-tuned compact LLMs for multilingual building automation in Chinese, English, and French. Their serverless architecture improves instruction interpretation and execution accuracy, though handling ambiguous and colloquial multilingual commands remains difficult.

An et al. (2024) present FunAudioLLM, a family of multilingual speech recognition and generation models with instruction-following capabilities, supporting zero-shot in-context learning and cross-lingual voice cloning. Their open-source models facilitate advanced multilingual voice interaction with LLMs.

Lu et al. (2025) propose Speech-IFeval, a framework to assess instruction-following in speech-aware language models and quantify catastrophic forgetting of textual skills. Their findings reveal significant gaps compared to text-based LLMs and sensitivity to prompt variations, underscoring the need for robust multimodal instruction-following evaluation.

Challenges and Insights in Multilingual Instruction-Following Evaluation

Performance Disparities and Resource Constraints

Multiple studies, including Marco-Bench-MIF (Zeng et al., 2025), XIFBench (Li et al., 2025), and Multi-IF (He et al., 2024), report substantial performance gaps between high-resource and low-resource languages in instruction-following tasks. These disparities are influenced by linguistic complexity, script differences, cultural factors, and data availability.

Khade et al. (2024) and Kulkarni et al. (2025) emphasize the difficulty of instruction-following evaluation in low-resource languages due to limited training data and evaluation resources, highlighting the need for native datasets and context-aware fine-tuning.

Instruction Complexity and Multi-Turn Interactions

AgentIF (Qi et al., 2025) and ComplexBench (Wen et al., 2024) reveal that LLMs struggle with lengthy, constraint-rich, and agentic instructions, particularly in managing multiple compositional constraints and tool specifications. Multi-IF (He et al., 2024) further shows that instruction adherence degrades over multi-turn conversations, especially in multilingual settings.

ReasonIF (Kwon et al., 2025) identifies significant failures of large reasoning models in following instructions during reasoning processes, with performance worsening as task difficulty increases. Instruction finetuning strategies improve adherence but substantial challenges remain.

Evaluation Methodology Limitations

Lyu et al. (2024) and Zeng et al. (2023) critique reliance on LLM-based evaluators due to biases and propose human response-guided composite evaluation frameworks and meta-evaluation benchmarks to increase reliability and alignment with human judgments.

Jourdan et al. (2025) find that conventional metrics like ROUGE inadequately capture instruction adherence in scientific text revision, advocating hybrid evaluation approaches combining LLM-based and domain-specific metrics.

Liu et al. (2025) highlight that instruction tuning can increase vulnerability to typographical noise, suggesting that robustness evaluation should be integral to instruction-following assessment.

Cross-Lingual Transfer and Alignment

Studies by Zhang et al. (2023), Bu et al. (2025), Shimabucoro et al. (2025), and Lai and Nissim (2024) demonstrate that cross-lingual transfer and alignment strategies, including pivot language use, multilingual representation alignment, and multilingual instruction tuning, are effective for enhancing multilingual instruction-following capabilities.

Zhang et al. (2023) and Ye et al. (2025) show that leveraging interactive translation and language separability can facilitate knowledge transfer and curriculum learning, improving instruction adherence across languages.

Sociocultural and Linguistic Nuances

Zhang et al. (2023) with SPARROW and Chitale et al. (2025) with Updesh emphasize the importance of sociopragmatic understanding and culturally grounded instruction data to improve instruction-following in diverse languages and contexts, particularly for low-resource and underrepresented varieties.

Maria (2025) and Zeng et al. (2025) underscore the need for domain and culture-specific adaptations in multilingual instruction-following, as seen in e-commerce and public health applications.

Conclusion

The evaluation of instruction-following capabilities in large language models has matured significantly, with a growing emphasis on multilingual contexts to ensure equitable and robust AI performance across languages. Foundational benchmarks like IFEval and their multilingual extensions (M-IFEval, Marco-Bench-MIF, XIFBench) provide standardized datasets and evaluation protocols that reveal substantial performance disparities tied to language resource levels, instruction complexity, and cultural factors.

Advances in multilingual instruction tuning, cross-lingual transfer, and representation alignment demonstrate promising strategies to enhance instruction adherence in low-resource and non-dominant languages. Domain-specific models

and multimodal LLMs further expand the scope of instruction-following evaluation, addressing real-world applications in public health, e-commerce, audio processing, and code generation.

Persistent challenges include managing complex, multi-turn instructions, ensuring robustness to noise and typographical errors, and developing reliable, human-aligned evaluation methodologies. Integrating sociocultural context and linguistic diversity into training and evaluation datasets remains crucial for advancing multilingual instruction-following.

Future work should continue to refine multilingual benchmarks with cultural and linguistic adaptations, improve cross-lingual transfer techniques, and foster robust, comprehensive evaluation frameworks that combine human and automated assessments. This will enable the development of LLMs capable of faithfully following instructions across the rich tapestry of human languages and applications.

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